PROGRAMMING FOR PROBLEM SOLVING (3110003)

Prepared By,

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Computer Engineering

LDCE, Ahmedabad

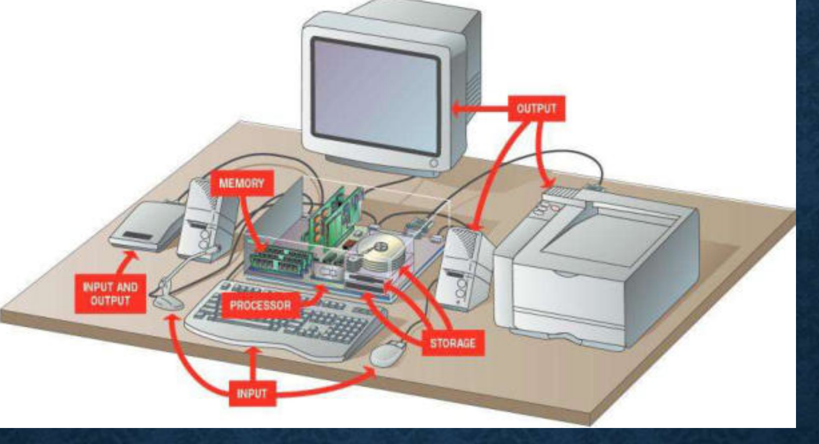
VISION

- To achieve academic excellence in Computer Engineering by providing value based education.

MISSION

- To produce graduates according to the needs of industry, government, society and scientific community.
- To develop partnership with industries, research and development organizations and government sectors for continuous improvement of faculties and students.
- To motivate students for participating in reputed conferences, workshops, seminars and technical events to make them technocrats and entrepreneurs.
- To enhance the ability of students to address the real life issues by applying technical expertise, human values and professional ethics.
- To inculcate habit of using free and open source software, latest technology and soft skills so that they become competent professionals.
- To encourage faculty members to upgrade their skills and qualification through training and higher studies at reputed universities.

Unit – 1
Introduction to Programming



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WHAT IS COMPUTER ?

- Computer is an electronic device used for storing, processing, and retrieving the information.

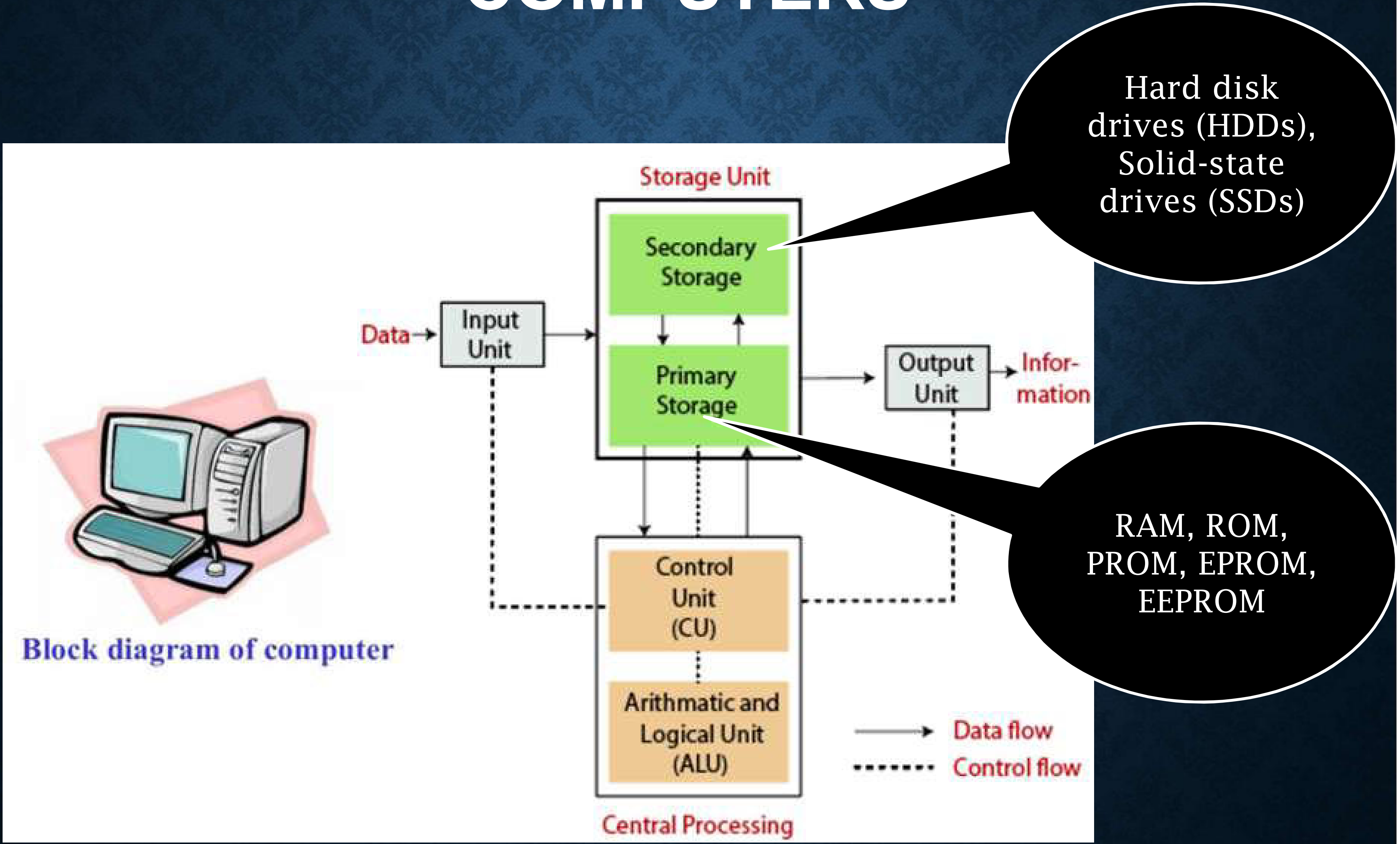
- Computer has mainly two components :
  - Hardware
  - Software.

DIFFERENCE BETWEEN HARDWARE & SOFTWARE

Hardware	Software
Physical component	Logical component
Can see and touch	Can not touch
It is divided into three parts : microprocessor , motherboard & Other peripherals.	It is divided into two categories: System Software & Application Software.
It is electronics components, so it is designed by the electronics engineer.	It is the collection of programs, so it is designed by computer engineer.
For maintenance, screw driver, tester, brush is required.	For maintenance, all types of software testing is required.
Ex : keyboard, mouse, monitor, CPU, printer………	Ex : Microsoft Office, Paint, loader, linker, Operating System……

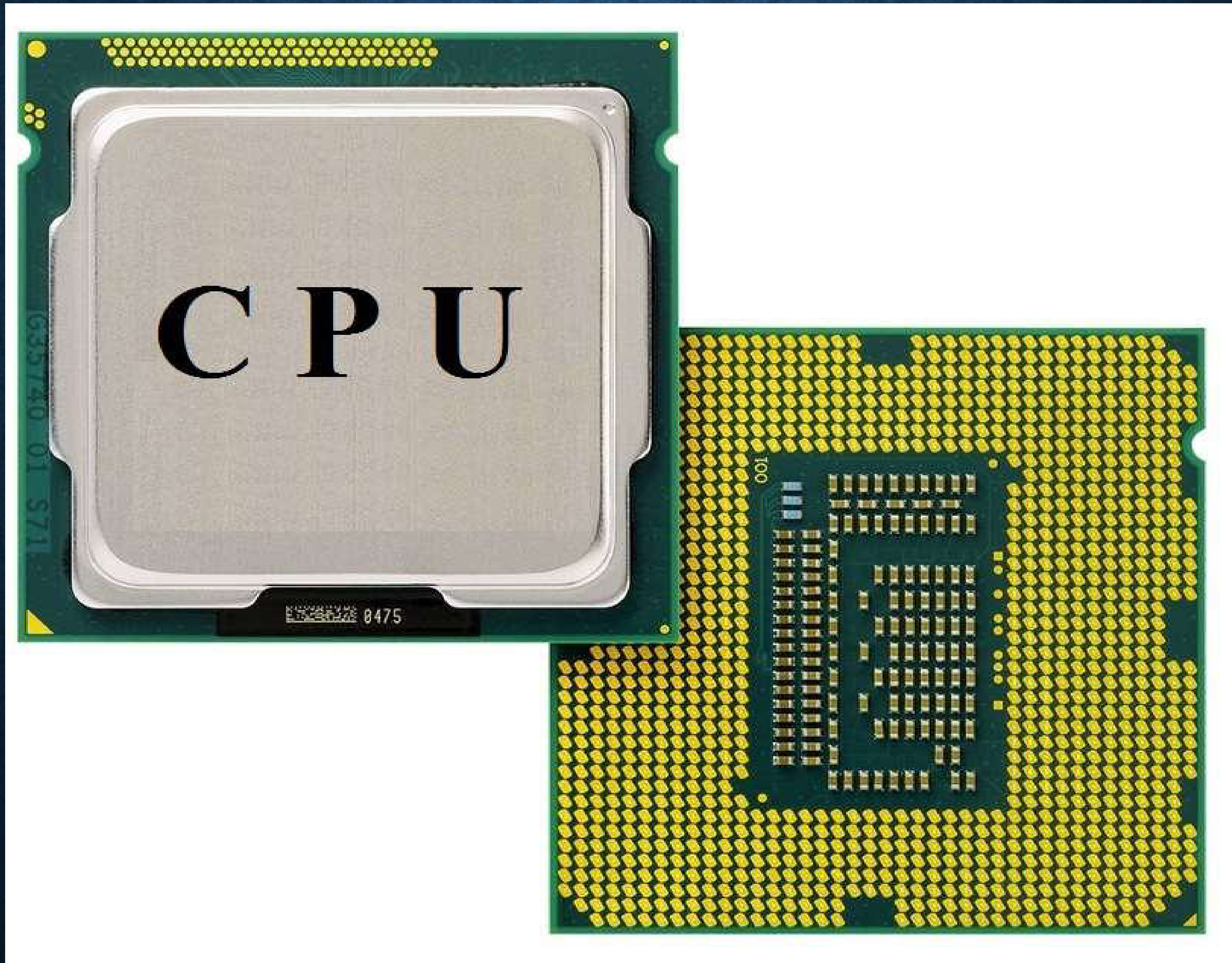
HARDWARE PARTS OF COMPUTER

BASIC BLOCK DIAGRAM OF COMPUTERS



Block diagram of computer

1. CENTRAL PROCESSING UNIT (CPU)



1. CENTRAL PROCESSING UNIT (CPU)

- Central Processing Unit is the brain of the computer. It is also known as a processor. The main purpose of the CPU is to execute the programs and stored in the memory.
- It consists main three parts: ALU, Main memory, C.U

1. ALU (Arithmetic Logical Unit) : It Performs Arithmetical Operations like addition, subtraction, multiplication, division and also performs logical operations like AND, OR, NOT operations.

1. CENTRAL PROCESSING UNIT (CPU)

2. C.U (Control Unit) : It controls all activities of C.P.U and other connected devices.

- Its main job is to fetch instructions from main memory and execute it over C.P.U.

- It has main two parts:

1. IR (Instruction Register) : It contains instructions which are in execution.
2. PC (Program Counter) : It contains the address of next instruction for execution.

1. CENTRAL PROCESSING UNIT (CPU)

3. Main memory : It is also known as primary memory or internal memory.
   It is divided in :

   1) RAM,

   2) ROM,

   3) PROM,

   4) EPROM,

   5) EEPROM.

1. Central Processing Unit (CPU)

- <u>RAM(Random Access Memory) :</u> It is volatile memory. It means the data will lost once the power is off.
- So it is also called temporary memory.
- The user can not store the data into RAM.

- <u>ROM (Read Only Memory) :</u> It is non-volatile memory. It means the data will retain even if the power is off.
- So it is also called permanent memory.

RAM

ROM

2. INPUT SECTION

- Input section of a computer consists of the devices which are used to enter the information into the computer from outside world.
- Example : Keyboard, Mouse, touch screen etc...

3.Output Section

- Output section of a computer consists of devices used to send the information to the outside world.
- The devices used for output are known as output devices.
- Example: Monitor, Printer, LCD, Speakers etc…

4. Secondary Memory

- It is also known as Auxiliary memory or external memory.
- This is permanent memory.
- User can stored data permanently on this memory.
- It can be modified.

- Ex : hard disk, floppy disk, CD, DVD, pen drive…etc.

Difference between Primary memory and secondary memory

Primary Memory	Secondary Memory
Store Current data	Store permanenet data
Less capacity	Large capacity
Faster	Slower
Volatile	Not volatile
Expensive	Chepaer

Advantages Of Computers

- The speed, accuracy, storage, reliability and automation are main advantages of computers.

1. **Speed:**

- The computer works at very high speed. So we can perform large program in a few seconds.
- Computer performs millions of operations in second.

2. **Accuracy:**

- There are many applications which need high degree of accuracy. For example, in space applicationthe path of the rocket is computed by computers without any mistakes.

Advantages Of Computers

3. Storage:

- Computers store large information using various storage devices available with computers.
- They not only store large information but also make them available when it is required for processing .

4. Reliability:

- Computer is a very reliable device.
- The information stored today in computer is available after years in same form.

Advantages Of Computers

5. Automation:

- Once the one task is created in computers, it can be performed by single click any time.
- For example, once the software for banking application is installed in computer, it calculates the interest quarterly or half yearly by just sending one command.
- The automation makes it possible to perform complex work by a click of command or button.

Limitations Of Computers

1. Lack of Intelligence:

- Humans have natural intelligence which makes them use common sense when any work in done. Computer can not think while doing work as it does not have natural intelligence.
- Computer only executes the instructions provided by human user.

Limitations Of Computers

2. Unable to correct mistakes:

- If human is provided the information which is incorrect, then it uses after correcting the information. Computer cannot correct the mistakes by itself.
- So if computer is provided wrong or incorrect information, it will use as it is. This makes the wrong work or wrong calculations.

Computer Software

- A software is a one type of program that contains sequence of instructions to perform certain functionality.

- There are two types of software :

1. System software
2. Application software

1.System Software

- System software is designed to operate the computer hardware.
- It is also provide platform for running application software.

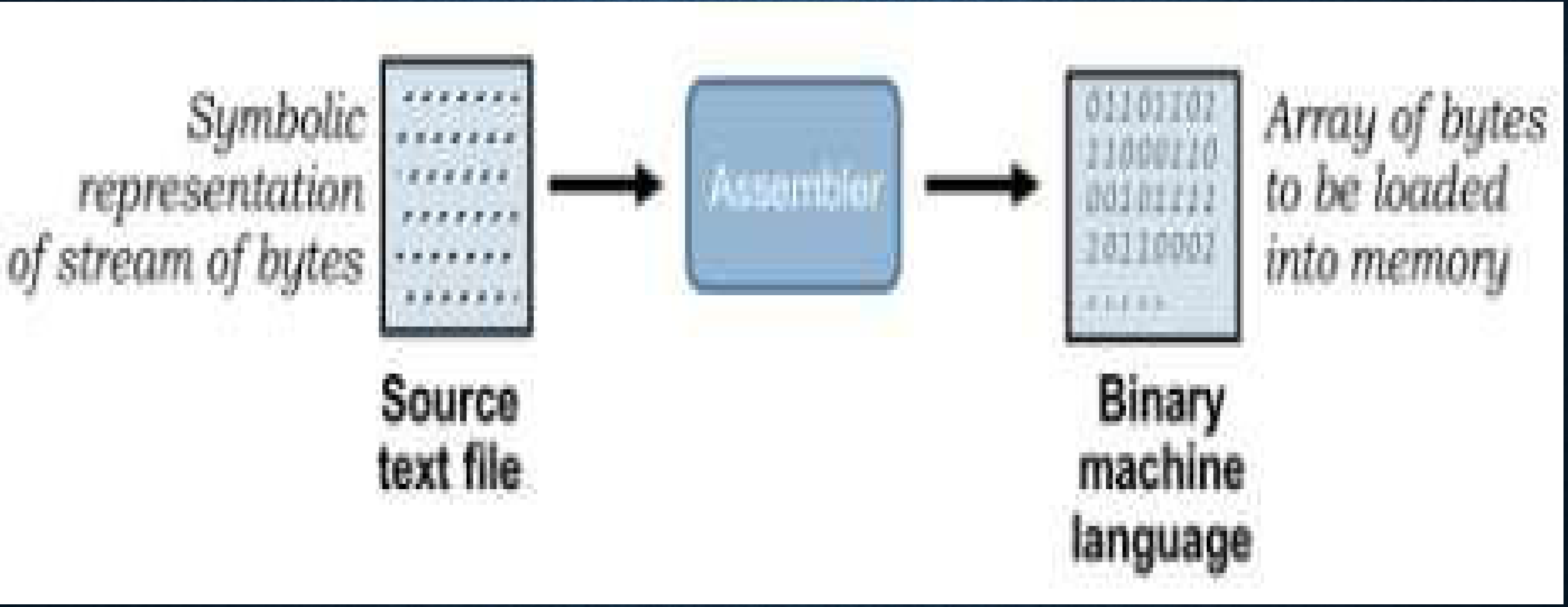
- Types of system software :

1. Operating system
2. Compiler
3. Interpreter
4. Assembler
5. Loader
6. Linker
7. Editor
8. Translator
9. Macro processor

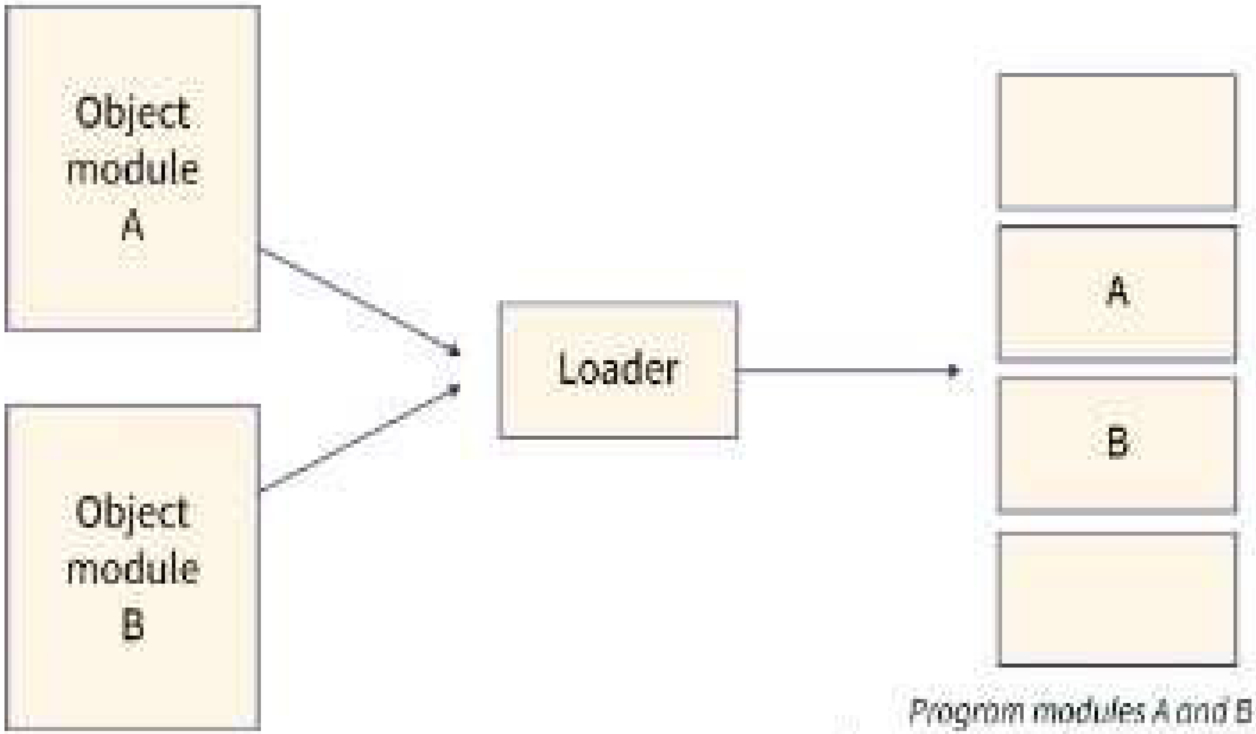
Difference between Compiler & Interpreter

	Compiler	Interpreter
1.	It checks the whole program at a time and then display the list of errors.	It checks the program line by line and stops checking whenever error occurs.
2	It converts whole source code into object code.	It converts source code into object code line by line.
3	After completion, source code is not required.	For, every execution of line, source code is required.
4	The object code of program generated when the program is error free.	It generates the object code for each line immediately if that line is error free.
5	The execution process of program is faster.	The execution process of program is slow.

1.SYSTEM SOFTWARE
ASSEMBLER

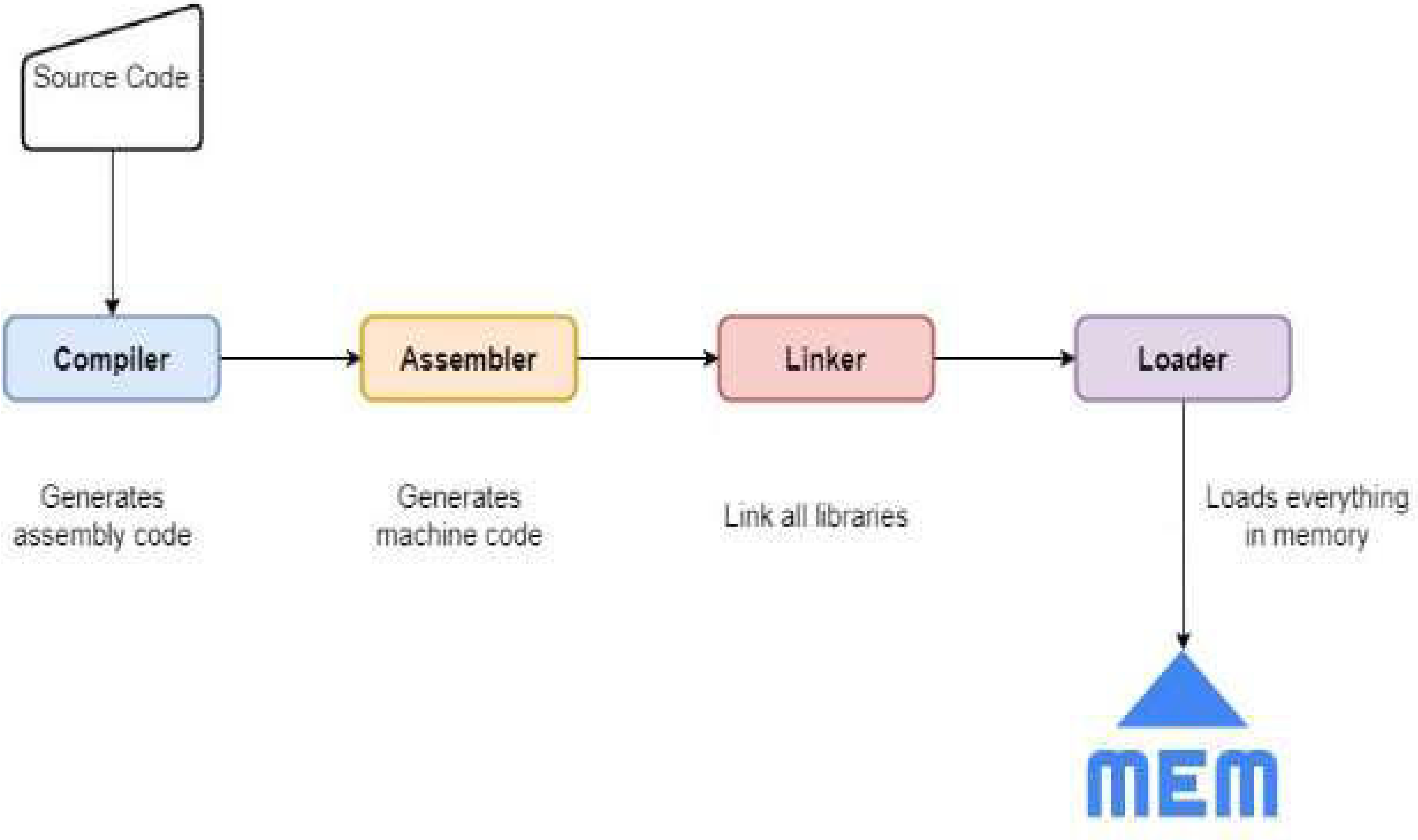


1.SYSTEM SOFTWARE
LOADER



Program modules A and B are loaded in the memory after linking. It is ready for execution

SYSTEM SOFTWARE



2.Application Software

- Application software is designed to perform specific user applications.

- There are two categories of application software :

1. General-purpose software : Microsoft Office, Oracle etc.

2. Specific-purpose software : tax calculation software, pay-roll system, Banking software's….etc

INTRODUCTION OF COMPUTER LANGUAGES

- Programming language is the language which is used to write the program.

- Computer languages are divided into two categories:
    1. Low level language
    2. High level language

- Low level language is divided into two types:
    1. Machine language
    2. Assembly language

PROGRAMMING LANGUAGES

1. Machine language programming :

- The languages written using binary language (0’s and 1’s) is know as Machine language.
- Computer can understand only low level language. So Computer don’t need to convert it, so low level languages are faster.
- Machine Level language is also known as Binary Language and Low Level language.

MACHINE LANGUAGE

- Advantages of Machine language programming:

1. The programs are compact and small.
2. The programs are stored in very less memory.
3. Once programs are entered into main memory computer can directly execute it without using any translator.
4. Suitable for low volume application.

MACHINE LANGUAGE

- Disadvantages of Machine language programming:

1. The programs are difficult to understand and debug.
2. It is required more time to write the program because it is written in bit form (0’s and 1’s)
3. The programmer often makes errors, which are difficult to Find.
4. The program can not describe because it is written in bits.
5. The programs are not portable.

PROGRAMMING LANGUAGES

2. Assembly language :

- A program written using set of instructions(mnemonics) is known as assembly language programming. (memory aid)

- Assembly language programming is also low level language.

- Computer requires assembler to translate assembly language into machine language.

ASSEMBLY LANGUAGE

- Advantages of Assembly language :

1. The program is easy to describe, debug & understand because of instructions.
2. Required less time to write the program.

ASSEMBLY LANGUAGE

- Disadvantages of Assembly language :

1. Required assembler to translate assembly language into machine language.
2. Programmer need to learn the structure of assembly language and syntax of every statements.
3. Assembly language programs are not portable because it is dependable on machine.

Programming languages

3. <u>Higher level language :</u>

- The languages written using English statements is known as high level language. Computer cannot understand high level language.
- So Computer have to convert high level language into low level language, so high level languages are slower than low level language.
- Higher level language is a potable language across machines.

HIGH LEVEL LANGUAGE

- Advantages of High level language :

1. The high level language is easy to understand or debug.

2. No need to learn computer architecture because it is not depend on the computer.

3. Required less time to write the program because it is written using English statements.

4. The programs are portable.

HIGH LEVEL LANGUAGE

- Disadvantages of High level language :

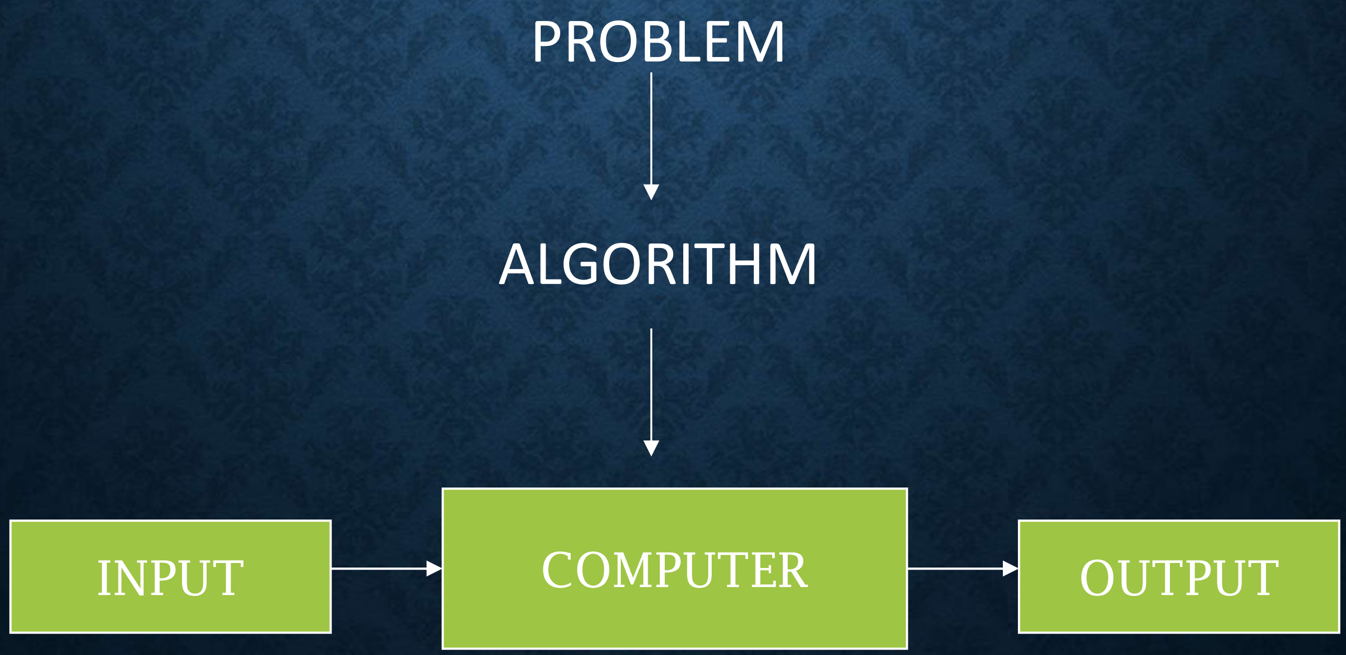
1. Required compiler or interpreter to translate high level language into machine level language.

2. Some high level languages are dependent on operating system.

ALGORITHMS AND FLOWCHARTS

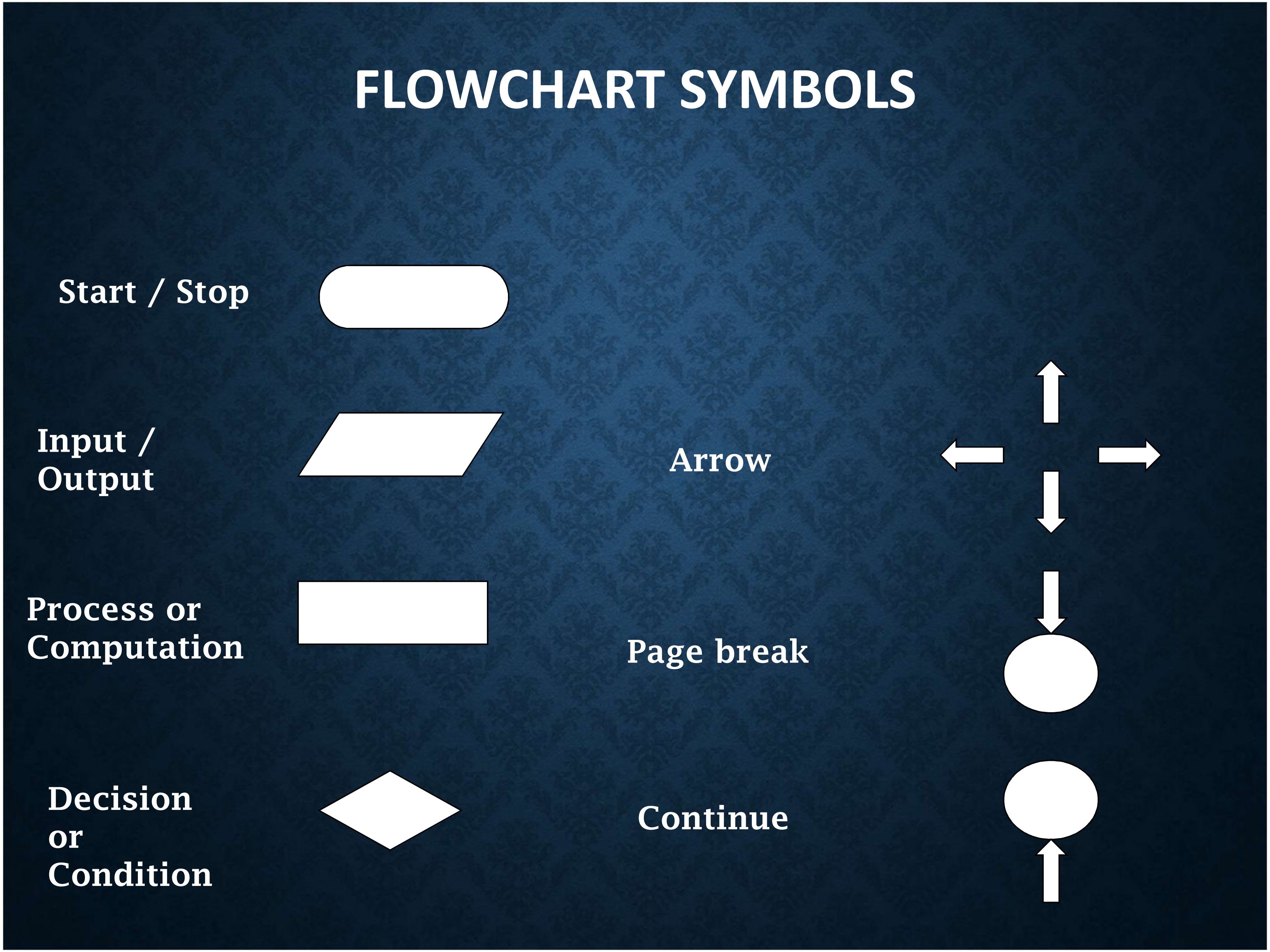
ALGORITHM

- An algorithm is a set of instructions to solve a problem .
- Basically an algorithm takes a set of values, as input and produces a value or set of values, as output.



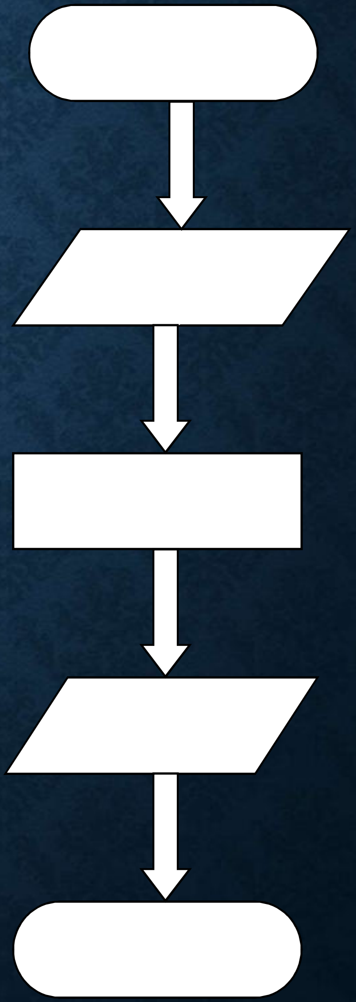
FLOWCHARTS

- A flowchart is a graphical representation of program that illustrates the sequence of operations to be performed to solve a given problem.


FLOWCHART SYMBOLS
Start / Stop
Input / Output
Process or Computation
Decision or Condition
Arrow
Page break
Continue

EXAMPLE1

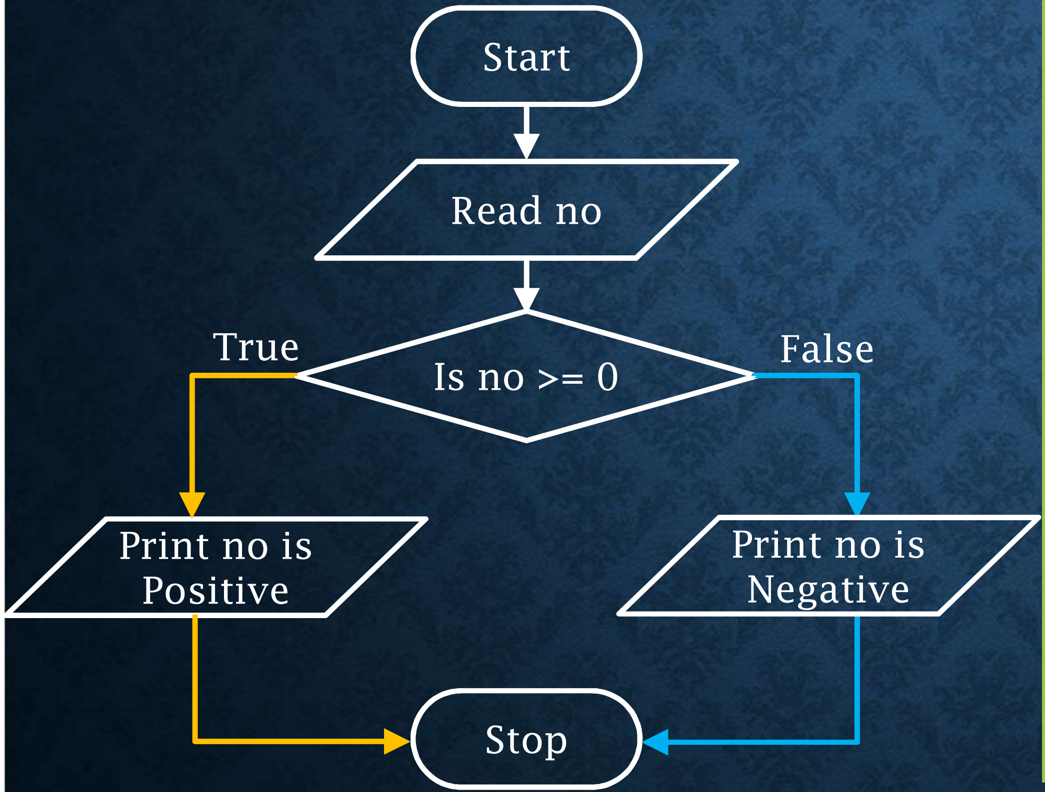
FLOWCHART



Algorithm

- Step 1: START
- Step 2: Input a ,b
- Step 3: Calculate SUM = a + b
- Step 4: PRINT SUM
- Step 5: STOP

EXAMPLE2



Step 1: Read no.

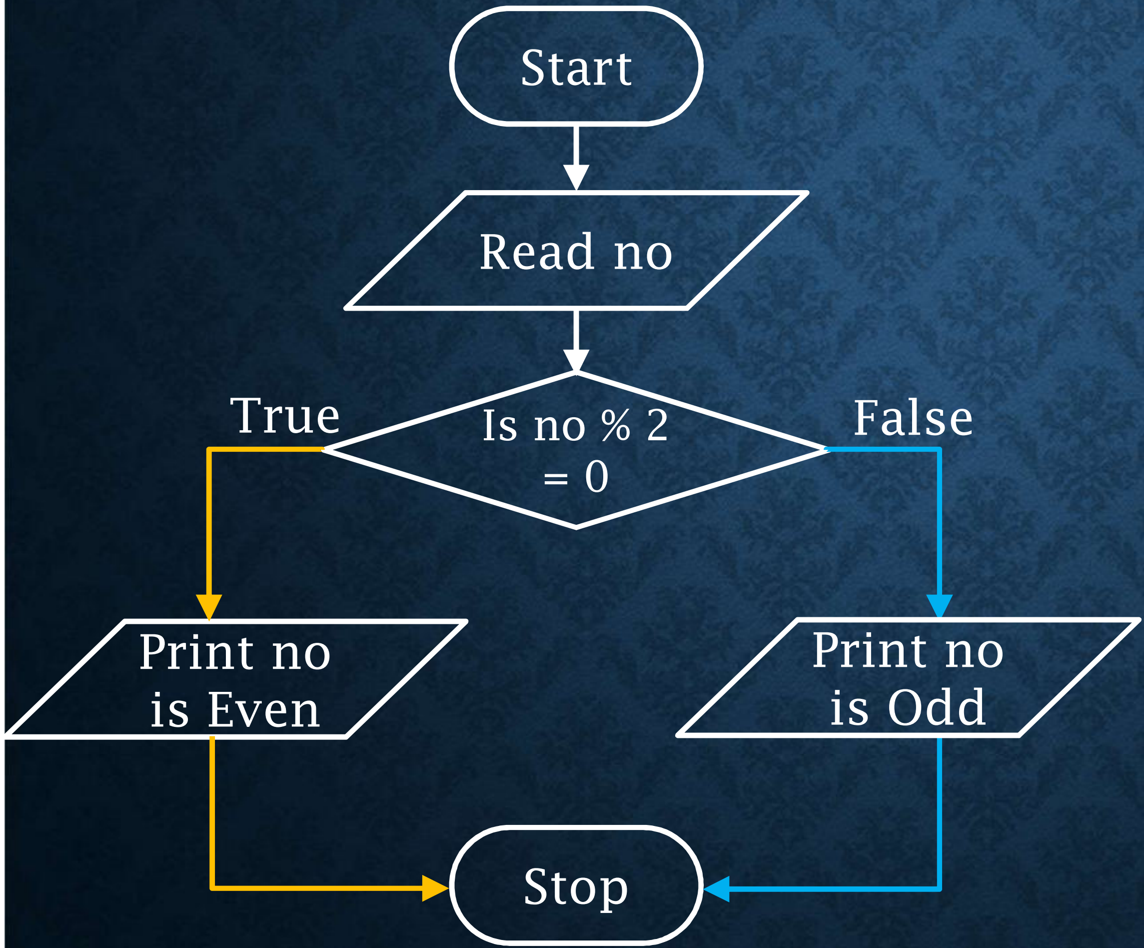
Step 2: If no is greater than equal zero, go to step 4.

Step 3: Print no is a negative number, go to step 5.

Step 4: Print no is a positive number.

Step 5: Stop.

EXAMPLE3



Step 1: Read no.

Step 2: If no mod 2 = 0,

go to step 4.

Step 3: Print no is a odd,

go to step 5.

Step 4: Print no is a even.

Step 5: Stop.

Thank You !